THE DERIVED TENSOR PRODUCT OF TWO PERFECT COMPLEXES IS PERFECT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

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Editorial note. The author writes 'the modules M^a \otimes_R L^b' where the symbol M refers to the first complex while the rest of the proof writes K^bullet; that is the author's text and is reproduced unchanged. The three references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Representation of a perfect object by a bounded complex of finite projective modules, and of the derived tensor product by a total complex. used at proof lines 19057-19061. Supplies the explicit complex whose terms are shown to be finite projective.
- Finite projective modules as direct summands of finite free modules, closed under tensor products. used at proof lines 19062-19065. Is the closure property that makes the total complex eligible.

Source reference. Original lemma 18889: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18889–18889. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 17066: sources/stacks-more-algebra/pinned-original-source.tex, source lines 17066–17066. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 16487: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16487–16487. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a ring. If K and L are perfect objects of $D(R)$, then $K \otimes^{\mathbf{L}}_R L$ is a perfect object too.*

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Proof. We can prove this using the definition as follows. We may represent K, resp. L by a bounded complex $K^\bullet$, resp. $L^\bullet$ of finite projective R-modules. Then $K \otimes^{\mathbf{L}}_R L$ is represented by the bounded complex $\mathrm{Tot}(K^\bullet \otimes_R L^\bullet)$. The terms of this complex are direct sums of the modules $M^a \otimes_R L^b$. Since M^a and L^b are direct summands of finite free R-modules, so is $M^a \otimes_R L^b$. Hence we conclude the terms of the complex $\mathrm{Tot}(K^\bullet \otimes_R L^\bullet)$ are finite projective.

Another proof can be given using the characterization of perfect complexes in Lemma 18889 and the corresponding lemmas for pseudo-coherent complexes (Lemma 16487) and for tor amplitude (Lemma 17066 used with $A = B = R$). $\square$